Amplitude Experiment Instrumentation Exercise



Implementing Experiment: Technical Exercise

This exercise accompanies the Partner Academy course IMPL-EXP Implement the Amplitude SDK.

Overview

This is a hands-on exercise for learners to gain experience implementing the Amplitude Experiment SDK in a simple web application. You'll use JavaScript; the exercise requires only a small number of edits to an existing application.

Participant Prerequisites

Each participant will need

- Text editor or a code editor (like VS Code)
- Access to your company's private Amplitude sandbox, including the following permissions:
    - Permission to retrieve API key for Analytics project
    - Permission to create a deployment in Amplitude Experiment
    - Permissions to create an experiment
- Copy of the SDK Implementation Assignment (Experiment Tech Track) Slide Template

Scenario

The product manager of the MemoMinder application has the idea that changing the label on the CTA to create a memo will increase the number of memos a user creates (and this increased engagement is a leading indicator of revenue). They want to test two different labels for the button that creates a new memo: "Add" and "Create" and see which one performs best on that metric.

Your assignment is to add the implementation of the Amplitude Experiment SDK to the MemoMinder web application, create the experiment in Amplitude Experiment and add the code to display the CTA text specified by the variant returned by the Experiment SDK for the experiment in question.

Procedure

Phase 1: Verify application and Analytics instrumentation are functioning.

1. Download and unpack sample code.
2. Look up the project's Analytics API key and replace AMPLITUDE_API_KEY with it in index.html. Note, there are 2 instances of AMPLITUDE_API_KEY.

   ACTION NEEDED: Drop one screenshot of your code that verifies step 1.2 into your copy of the SDK Implementation Assignment (Experiment Tech Track) Slide Template**.**

3. Bring up the MemoMinder app by loading the local file index.html in a browser.
4. Send a minimum of three events to Amplitude by creating, closing, and deleting memos.
5. Verify events are being received from your instance of the application in the Analytics project with one of the following:
   a. User Profiles
   b. Amplitude Event Explorer Chrome Extension
   c. Event Explorer in Amplitude Analytics

   ACTION NEEDED: Take one screenshot that verifies a minimum of three *Memo Interacted* events have been received by Amplitude, as described in step 1.5. Drop it into your assignment slide deck.

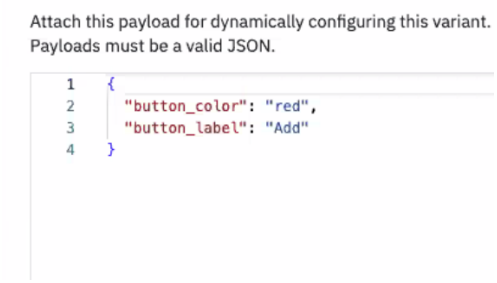
Phase 2: Create deployment, create, configure, and test experiment

1. In Amplitude Experiment, create a Client-side deployment associated with your Analytics project.

   ACTION NEEDED: Take one screenshot of Amplitude Experiment that verifies step 2.1. Drop it into your assignment slide deck.

2. Create an experiment named *team-name-memominder-label* with settings:
   a. A/B test
   b. Remote evaluation
   c. Bucketed by Amplitude ID
   d. Added to the deployment you created above
3. Configure that experiment to have:

a. Two variants: add and create
b. A JSON payload for each variant with (at a minimum) a field named `button_label` with the values (respectively) "`Add`" and "`Create`".



ACTION NEEDED: Take one screenshot of Amplitude Experiment that verifies the variants are configured as described in steps 2.3.a and 2.3.b. Drop it into your assignment slide deck.

c. Allocation to *All Users* at a 50/50 ratio between the two variants, rolling out to 100%.
d. Primary metric measured as *Event Totals* with the event being *Memo Interacted where action = add.*

ACTION NEEDED: Take one screenshot of Amplitude Experiment that verifies the experiment is configured as described in steps 2.3.c and 2.3.d. Drop it into your assignment slide deck.

4. Activate the experiment
5. Use the web interface to the Evaluation API to verify that the experiment is ready to assign users. You will need values for:
   a. **Deployment_key** (key from the deployment created in step 2.1)
   b. **device_id** (use whatever random string you'd like)
   c. **Flag_key** (the key you specified in step 2.2)
6. Check that you are receiving a key and payload field as expected.

ACTION NEEDED: Take one screenshot of the Evaluation API page that verifies step 2.6. Drop it into your assignment slide deck.

7. Check *User Profiles* for the associated Analytics project and verify an Assignment event was received.

ACTION NEEDED: Take one screenshot of Amplitude that verifies step 2.7. Drop it into your assignment slide deck.

Phase 3: Instrument the application to implement the experiment

1. Edit the file index.html to add the Script tags to load and initialize the Experiment SDK to the file index.html.
2. Replace the string DEPLOYMENT_KEY with the deployment key from the deployment you created in step 2-1.

   ACTION NEEDED: Take one screenshot of your code that verifies step 3.1 and 3.2. Drop it into your assignment slide deck.

3. Edit the definition of the function setLabel to use methods from the Experiment SDK to set the label of the add-memo button to the value of the button_label field of the JSON payload associated with the variant assigned by Experiment to the current user.
   a. This will involve calls to both start() and variant().
      Hint: fetch() returns a promise that resolves when the resulting HTTP call has returned and the SDK has cached all of the assignments for the current user. Until that promise has resolved, calls to variant() may return null. start() triggers fetch() automatically.

   ACTION NEEDED: Take one screenshot of your code that verifies step 3.3. Drop it into your assignment slide deck.

4. If you are stuck and need help with your code, email kelly.reed@amplitude.com for a full working version.

Phase 4: Verify Experiment instrumentation is working correctly

1. Reload your browser with your updated version of index.html
2. See that the button has a label of either “Add” or “Create”.

   ACTION NEEDED: Take one screenshot of your MemoMinder app that verifies step 4.2. Drop it into your assignment slide deck.

3. Create/Add a memo to the list
4. Check *User Profiles* to verify the following events have been triggered (from the same user):

a. An Assignment event
b. An Exposure event
c. A Memo Interacted event

ACTION NEEDED: Take one screenshot of your MemoMinder app that verifies step 4.4. Drop it into your assignment slide deck.

5. Load the page in a new incognito window. Refresh until you see a different button label than what you saw in step 4.2.

   ACTION NEEDED: Take one screenshot of your MemoMinder app that verifies step 4.5. Drop it into your assignment slide deck.

6. Examine the information for the experiment in the *Activity* tab in Experiment and verify that users have been added to the totals for *Assignments* and *Exposures*.

   ACTION NEEDED: Take one screenshot of Amplitude Experiment that verifies step 4.6. Drop it into your assignment slide deck.